

KWESI A. QUAGRAINE

Chief Scientist and climate researcher specializing in climate impacts, solar radiation modification (SRM), and climate intervention governance, with experience bridging scientific research, multilateral engagement and climate policy for vulnerable countries.

- Expertise in Earth system modeling, climate impacts, SRM research, and regional climate risk assessment
- Leadership of international science-policy initiatives focused on climate vulnerability and emerging climate interventions
- Experience supporting ministerial, parliamentary, and multilateral climate engagement processes
- Strong record of translating complex scientific evidence into policy-relevant analysis and strategic communications

EDUCATION

PH. D., CLIMATOLOGY – University of Cape Town, Cape Town, South Africa 2021
Dissertation: Dynamics of Co-Behavior of Climate Processes over Southern Africa

B.S., & MPhil PHYSICS – University of Cape Coast, Cape Coast, Ghana 2011 & 2014

Minors: METEOROLOGY

Thesis: West African Monsoon (WAM) Jump and Its Implication for Rain-fed Agriculture

EXPERIENCE

CHIEF SCIENTIST – Climate Vulnerable Forum-Vulnerable Twenty (CVF-V20) Group 2026 – date

- Leading the Science and Health agenda for the 1.7 billion climate vulnerable people.

POSTDOCTORAL RESEARCH FELLOW – National Center for Atmospheric Research, Boulder 2023 – 2026

- Designed a workflow to regionally refine CESM-SAI models using variable resolution CESM for impact assessments.
- Understanding the impacts associated with Stratospheric Aerosol Injection (SAI).
- Developed high-pressure system indices for climate variability studies.

LECTURER & SENIOR LECTURER – University of Cape Coast, Ghana 2020 – 2023

- Designed courses and taught numerous courses in atmospheric physics and meteorology.
- Consulted with International organizations on climate outputs.

POSTDOCTORAL RESEARCH FELLOW – University of Cape Town, South Africa 2020 – 2023

- Designed a workflow for understanding the behavior of large-scale climate drivers in modulating the regional climate.
- Consulted with International organizations on climate outputs.

RESEARCH SCHOLARSHIPS & GRANTS

Reflective High Latitude Low Altitude (HiLLA) Research Grant 2025 – 2026

- Investigate the impact of HiLLA on the West African Monsoon and its dynamics (Co-Investigator).

The DEGREES Solar Radiation Management Governance Initiative: Socio-Political Research Fund 2024 – 2027

- Assessment of knowledge and governance towards Solar Radiation Modification across Ghana: Implications for more efficient Ethical, Policy and Social Practices (Co-Investigator).

The DEGREES Solar Radiation Management Governance Initiative: Modeling Research Fund 2023 – 2027

- Solar Radiation Management with Stratospheric Aerosol interactions with Harmattan and associated impacts on climate variables over Southern West Africa (Côte d'Ivoire, Ghana, Togo, Benin, Nigeria, Cameroon) (Co-Investigator).

- National Research Foundation (NRF) SA Research Chair in Climate Change (SARCHI) Scholarship** 2016 – 2023
- Investigating the combined influence of large-scale climate processes in driving regional climate.
- Future Resilience for African Cities And Lands (FRACTAL), UK DFID** 2016 – 2021
- Advance scientific knowledge about regional climate responses to human activities and work with decision-makers to integrate this knowledge into climate-sensitive decisions at a city & regional scale (Worked on the regional climate understanding package).
- Newton Research Fellow, Climatic Research Unit, University of East Anglia, UK** 2017

CONSULTANCIES

- UNEP-Green Climate Fund & Ghana National Adaptation Plan (EPA, Ghana)** 2016 – 2021
- Enhancing multi-sector planning and capacity for effective adaptation in Ghana.
- User Interface Project (UIP) for the World Meteorological Organization (WMO)** 2017
- Developing of a UIP framework for worldwide use in climate services.
- Regional Climate Projection Information for ICLEI Local Government for Sustainability** 2021 – 2022
- Development of climate change projections for Bo City, Sierra Leone and Cape Coast, Ghana.

AWARDS AND HONORS

- NCAR CISL Allocation Recipient | NCAR, Boulder, CO** 2016 – date
- Supercomputing resources to improve CESM-SAI models using regional refinement for impact analysis.
- International/Refugee Grant | University of Cape Town, South Africa** 2017 – 2021

PUBLICATIONS

SELECTED JOURNAL ARTICLES. (Complete publication list available at scholar.google.com/kaquagrain)

- Quagraine, K. A., et al.** (2025). *Africa must lead the governance on solar radiation management*. Nature Climate Change.
- Quagraine, K. A., et al.** (2025). *Assessing the impact of Stratospheric Aerosol Injection on precipitation extremes in Africa using the ARISE-SAI-1.5 dataset*. Environmental Research: Climate.
- Quagraine, K. A., et al.** (2020). *Using co-behavior analysis to interrogate CMIP5 GCM performance over Southern Africa*. Journal of Climate.
- Quagraine, K. T., Quagraine, K. A., et al.** (2025). *Assessing the impact of the South Indian Ocean High-Pressure system using a novel index*. International Journal of Climatology.
- Nkrumah, F., Quagraine, K. A., et al.** (2025). *Assessing regional climate trends in West Africa under geoengineering: A multi-model comparison of UKESM1 and CESM2*. Journal of Geophysical Research - Atmospheres.
- Richter, J. H., ... Quagraine, K. A., et al.** (2026). *Earth system predictability across time scales for a resilient society*. BAMS.
- Quagraine, K. A., et al.** (2024). *A simple subtropical high-pressure system index over the South Atlantic*. Atmospheric Science Letters.
- Quagraine, K. A., et al.** (2019). *A methodological approach to assess the co-behavior of climate processes over Southern Africa*.

VOLUNTEER / LEADERSHIP ACTIVITIES

- Recycle Up! Ghana | Technik Ohne Grenzen e. V., Germany** 2016 – date
- In conjunction with Technik Ohne Grenzen, I have trained the youth in Ghana on the benefits of reducing, reusing and recycling plastics for the benefit of the climate.
- Notable Outcomes:
- 6 Summer Camps in Accra, Cape Coast and Kumasi, Ghana.
 - Lead training workshops on recycling
 - 10 Established recycling youth groups in Senior High Schools across Ghana.

- Teaching Computer Basics | Technik Ohne Grenzen e. V., Germany** 2016 – date
- Notable Outcomes:
- 2 Schools resourced with a Computer Laboratory for improved learning.